

Hiroataka Aihara

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EDUCATION

The University of California, Berkeley | B.S. in Data Science

Aug. 2023 - Present

Relevant Coursework: Data Structures & Algorithms, Principles & Techniques of Data Science, Probability for Data Science, Linear Algebra, Deep Learning, Intro to Machine Learning, Intro to AI

WORK EXPERIENCE

Credo | *Software Engineering Intern*

May 2026 - Present

- Designed an end-to-end Python workflow engine with an HTML ticketing-system interface, using workNode and switchNode architectures to streamline 10,000+ part numbers and 1,000,000+ testing records company-wide
- Mined an Excel dataset of 10,000+ part numbers to identify existing naming patterns, designing a standardized part number naming convention for company-wide use that keeps part tracking consistent across all workflows
- Deployed a Django app to production over HTTPS with Nginx and Gunicorn, pulling live testing data from PDM via switchNode into MongoDB, and applied neural network models to classify whether test results meet target metrics

Whitney Laboratory | *Undergraduate Research Assistant*

Sep. 2025 - Present

- Designed a controlled experiment standardizing pupil measurements for 200+ participants across 1,000+ trials, building an automated preprocessing pipeline with z-score normalization that reduced inter-subject variability by 67%
- Ran statistical analysis (linear regression, $R^2=0.43$, $p<0.001$) across 3,000+ trials to test pupil-accuracy relationships, finding 23% higher task performance during high-attention periods and validating engagement hypotheses

Lamu | *Data Engineer Intern*

Jan. 2026 - May 2026

- Designed AWS cloud infrastructure (Lambda, PostgreSQL, QuickSight) tracking 15+ KPIs across distributed services, delivering visualization dashboards and insights to product stakeholders that improved registration rates
- Engineered ETL pipeline processing 500K+ user records from DynamoDB to PostgreSQL, writing SQL queries to track cohort retention and user growth trends that guided app and website improvements

BLCK UNICRN | *Software Engineer Intern*

Sep. 2025 - Dec. 2025

- Built Python ETL pipelines processing 50,000+ prompt-response pairs to construct and validate LLM training datasets for an internal company chatbot, achieving 95% data quality through automated validation checks
- Engineered an automated anomaly detection framework with 12+ checks to flag data quality issues across LLM training datasets, reducing dataset errors by 78% and improving chatbot response reliability for internal teams
- Designed MCP configuration architecture using JSON schemas to govern LLM behavior and response patterns across 8 prompt categories, improving response consistency by 45% through structured data governance

Codify | *Software Developer*

Sep. 2023 - Sep. 2024

- Delivered frontend work for 3 enterprise client projects including Intuit, refactoring render-blocking JavaScript and removing 15+ unused script dependencies to cut Intuit's homepage load time from ~10s to ~5s

PROJECTS

Spam/Ham Email Classifier | *Python, NLP, Scikit-learn, TF-IDF, Random Forest*

- Built a binary NLP classifier to detect spam versus legitimate emails, applying tokenization, stop word removal, punctuation stripping, and TF-IDF vectorization to reduce noise in raw text data by 40% before model training
- Trained and tuned a Random Forest classifier with hyperparameter search, achieving 92.3% accuracy and a 0.90 F1 score, balancing precision and recall to limit both missed spam and false positives on legitimate email

Touch Grass | *Python, Ollama, Qdrant, RAG, Google Calendar API, Vector Database*

- Built a full-stack AI digital wellness coach in 36 hours with a team of 2, using retrieval-augmented generation (RAG) with Ollama and Qdrant to retrieve relevant past interventions and generate personalized recommendations
- Designed prompt engineering strategies to generate context-aware interventions based on user habits and behavioral history, then validated retrieval performance against 1,000+ AI-generated synthetic user profiles

SKILLS

- Data Science & ML:** Python, PyTorch, Scikit-Learn, NLP, LLMs, RAG, Deep Learning, Classification, Anomaly Detection, Data Mining, Model Context Protocol (MCP), Agent Systems, Statistical Analysis
- Software Engineering:** Java, JavaScript, TypeScript, HTML/CSS, Django, Flask, Nginx, Gunicorn, REST APIs, AWS (Lambda, RDS, S3), PostgreSQL, MongoDB, DynamoDB, Apache Airflow, Git, CI/CD
- Languages:** English, Chinese, Japanese